AP PHYSICS 1

Unit 1 • Worksheet 1.3A • Answer key

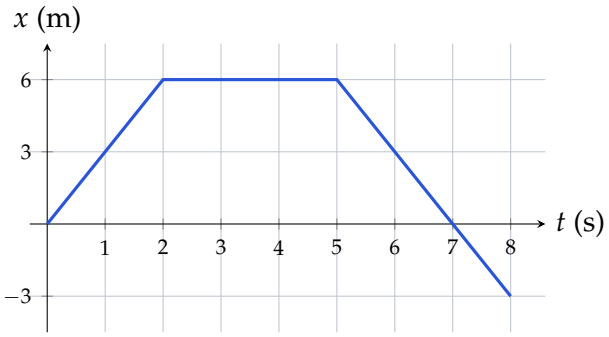
Worksheet 1.3A • Representing Motion

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Questions 1 to 4 refer to the graph below: the position-time graph of an object moving along a line.



Question 1

What is the velocity of the object between $t = 0$ and $t = 2\,\text{s}$?

(A) $6\,\text{m/s}$
(B) $3\,\text{m/s}$
(C) $2\,\text{m/s}$
(D) 0

Solution

Answer: (B). Slope $= 6/2 = 3\,\text{m/s}$.

Question 2

Between $t = 2\,\text{s}$ and $t = 5\,\text{s}$, the object is

(A) at rest
(B) moving at a constant positive velocity
(C) speeding up
(D) moving backward

Solution

Answer: (A). The position stays at 6 m: the slope is zero, so the velocity is zero.

Question 3

What is the velocity of the object between $t = 5\,\text{s}$ and $t = 8\,\text{s}$?

(A) $3\,\text{m/s}$
(B) $-9\,\text{m/s}$
(C) $-3\,\text{m/s}$
(D) $1\,\text{m/s}$

Solution
Answer: (C). Slope $= (-3-6)/(8-5) = -3\,\text{m/s}$: it moves in the negative direction.

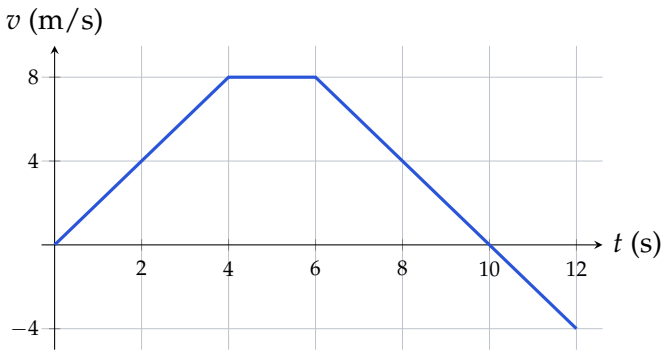
Question 4

What is the displacement of the object from $t = 0$ to $t = 8\,\text{s}$?

(A) $6\,\text{m}$
(B) $-3\,\text{m}$
(C) $15\,\text{m}$
(D) $3\,\text{m}$

Solution
Answer: (B). $\Delta x = x(8) - x(0) = -3 - 0 = -3\,\text{m}$.

Questions 5 to 8 refer to the graph below: the velocity-time graph of a cart moving along a line.



Question 5

What is the acceleration of the cart at $t = 2\,\text{s}$?

(A) $8\,\text{m/s}^2$
(B) $4\,\text{m/s}^2$
(C) $2\,\text{m/s}^2$
(D) 0

Solution
Answer: (C). Slope from 0 to 4 s: $8/4 = 2\,\text{m/s}^2$.

Question 6

What is the displacement of the cart from $t = 0$ to $t = 6\,\text{s}$?

(A) $32\,\text{m}$
(B) $48\,\text{m}$
(C) $16\,\text{m}$
(D) $24\,\text{m}$

Solution
Answer: (A). Area: triangle $\frac{1}{2}(4)(8) = 16\,\text{m}$ plus rectangle $2 \times 8 = 16\,\text{m}$: $32\,\text{m}$.

Question 7

At what time does the cart change direction?

(A) $6\,\text{s}$
(B) $4\,\text{s}$
(C) $12\,\text{s}$
(D) $10\,\text{s}$

Solution
Answer: (D). The velocity changes sign where the graph crosses the t-axis: at $t = 10\,\text{s}$.

Question 8

What is the displacement of the cart from $t = 0$ to $t = 12\,\text{s}$?

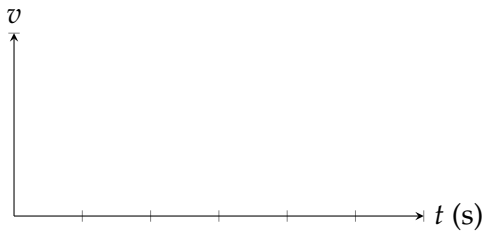
(A) $52\,\text{m}$
(B) $48\,\text{m}$
(C) $44\,\text{m}$
(D) $40\,\text{m}$

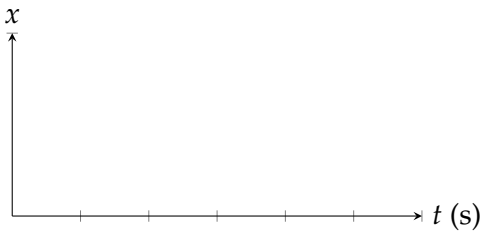
Solution
Answer: (C). Areas: $16 + 16 + 16$ above the axis (to $t = 10\,\text{s}$), then $-\frac{1}{2}(2)(4) = -4\,\text{m}$ below: $48 - 4 = 44\,\text{m}$. (The distance is $52\,\text{m}$.)

Question 9 (Free response)

A cart starts from rest at $x = 0$. It accelerates at $2\,\text{m/s}^2$ for $3\,\text{s}$, then moves at constant velocity for $3\,\text{s}$.

(a) On the first axes, sketch the velocity-time graph from $t = 0$ to $t = 6\,\text{s}$. Label the values.
(b) On the second axes, sketch the position-time graph. Describe the shape of each part.
(c) Find the displacement at $t = 6\,\text{s}$.





Solution

(a) A straight line from $(0, 0)$ to $(3\,\text{s}, 6\,\text{m/s})$, then horizontal at $6\,\text{m/s}$. (b) A parabola opening up (slope increasing) from 0 to $3\,\text{s}$, reaching $9\,\text{m}$, then a straight line of slope $6\,\text{m/s}$. (c) $\frac{1}{2}(3)(6) + 3(6) = 9 + 18 = 27\,\text{m}$.